Exp: 2

STUDY OF DIFFERENT TYPES OF NETWORK CABLES.

Aim:

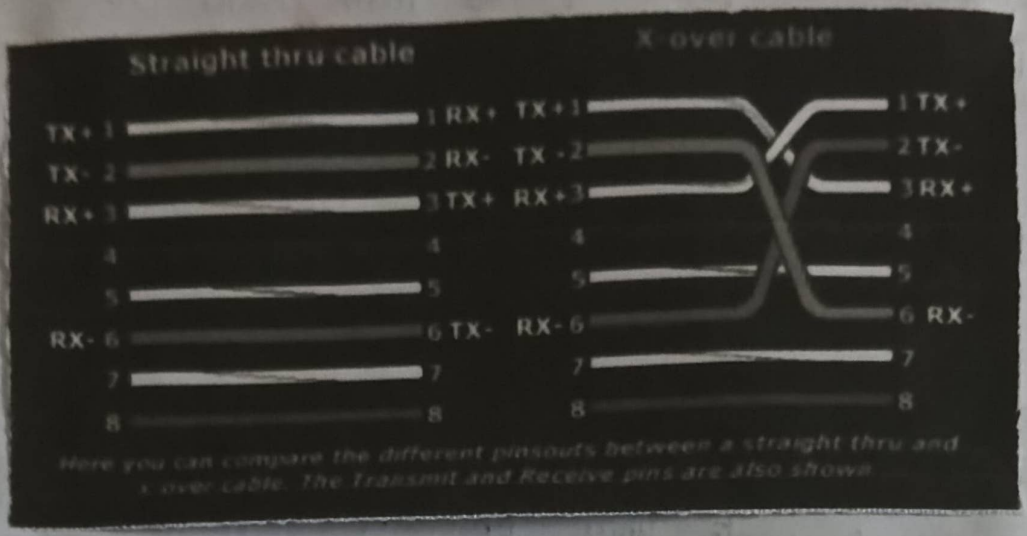
To Study of different types of Network cables.

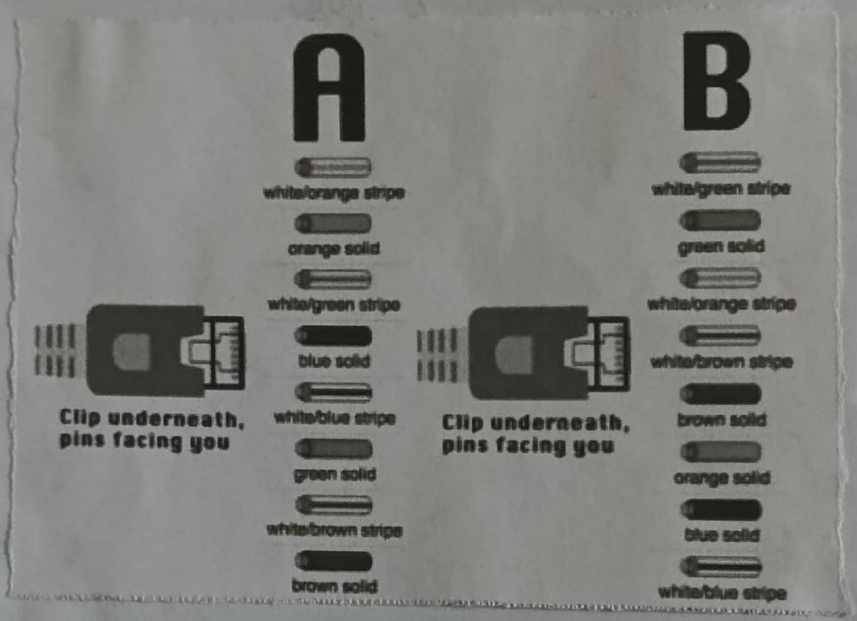
a) Understand different types of Network cable.

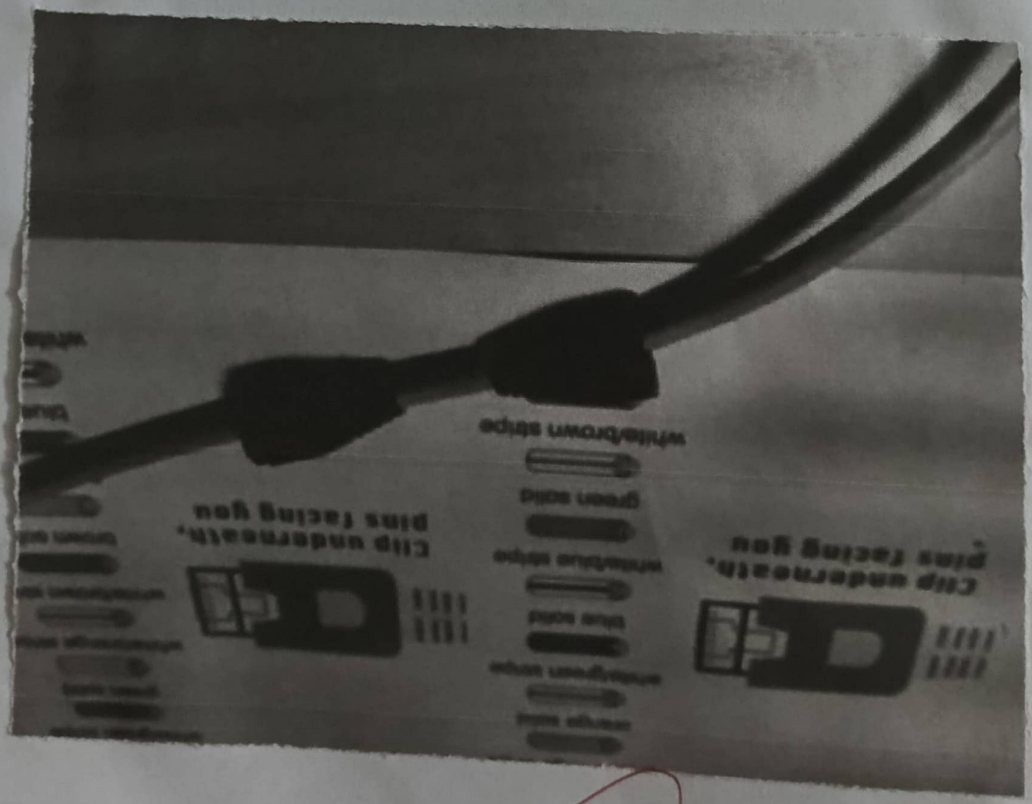
Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial cable
4. Fibre optic cable

Cable Type	category	Maximum Data Transmission	Advantages / Disadvantages	Application / use
UTP	Category 3	10bps	Advantages * Cheaper in cost * Easy to install as they have a smaller overall diameter Disadvantage * More prone to EMI	10 Base-T Ethernet
	Category 5	Upto to 100 Mbps		Fast Ethernet gigabit Ethernet
	Categoryse	1Gbps.		Fast Ethernet, gigabit Ethernet
STP	Category 6, 6a	10Gbps	Advantage * Shielded. * Faster than UTP * Less susceptible to noise and Interfere Disadvantage * Expensive * greater installation effort	gigabit Ethernet 10G Ethernet (55 m) widely used in data centres
SSTP	Category 7	10Gbps		gigabit Ethernet 10G Ethernet (100m)


Straight thru cable
X-over cable
TX+ 1
TX- 2
RX+ 3
4
5
RX- 6
7
8
1 RX+
2 RX-
3 TX+
4
5
6 TX-
7
8
TX+ 1
TX- 2
RX+ 3
4
5
RX- 6
7
8
1 TX+
2 TX-
3 RX+
4
5
6 RX-
7
8
Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also showa


A
white/orange stripe
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
brown solid
Clip underneath, pins facing you
B
white/green stripe
green solid
white/orange stripe
white/brown stripe
brown solid
orange solid
blue solid
white/blue stripe
Clip underneath, pins facing you


white/brown stripe
green solid
white/blue stripe
blue solid
white/green stripe
orange solid
Clip underneath, pins facing you
Clip underneath, pins facing you

Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	* High bandwidth * Low loss bandwidth * versatile Disadvantage * Limited distance * cost	Speed of singral is 500 m Television network High Speed internet connection
Fibre optics cable	Single mode Multi mode	100 Gbps	Advantages * High Speed * High security * Long distance Disadvantages * Expensive.	Maximum distance of fibre optics cable is around 100 meter

b) Make Your Own Ethernet Cross-Over cable/ Straight cable

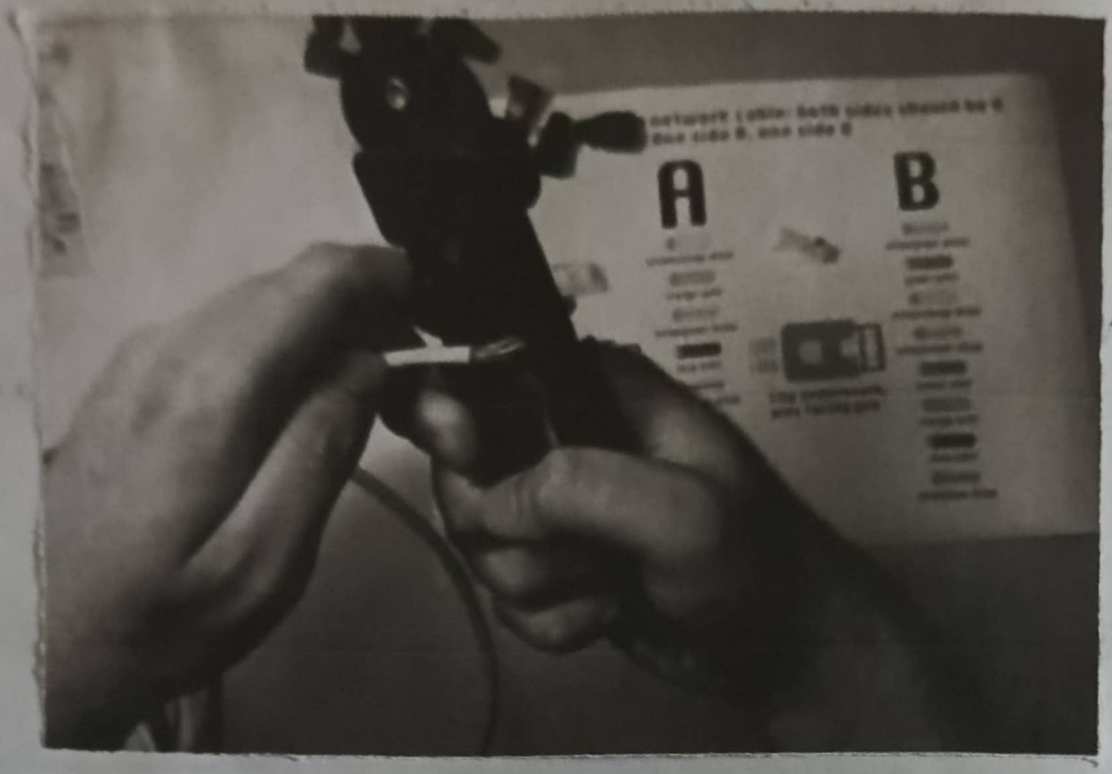
Tools and parts needed:

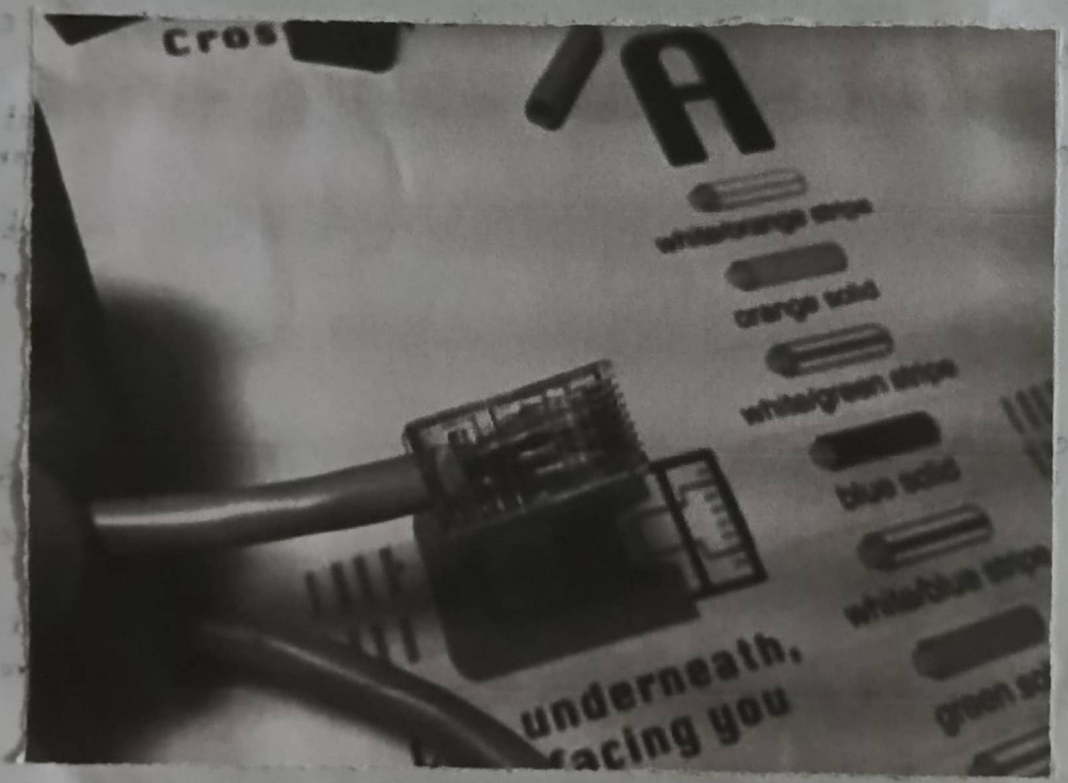
* Ethernet cabling. CAT5e is certified for gigabit support. but CAT5 cabling works as well, just over shorter distances.
* A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
* Two RJ45 plugs.
* Optional two plug shields.

Step 1: To start construction of the device, begin by threading sheilds onto the cable

Step 2: Next, strip approximately 1.5 cm of cable sheilding from both ends. The crimping tools has a round area to complete this task

Step 3: After, you will need to untangle the wire; there should be four "twisted pairs". Referencing back to the sheet. arrange them from top to bottom. one end should be in arrangement A and other in B


A
B


A
underneath,
facing you

Step 4: Once the order is correct, bunch them together in line and if there are any that stick out farther than others, snip them back to create an even level. To do so, hold the plug with the clip side facing away from you and have the gold pins facing towards you, as shown.

Step 5: Next, push the cable right in, the notch at the end of the plug needs to be just over the cable sheildings

Step 6: After the wires are securely sitting inside the plug insert it into the crimping tool and push down.

Step 7: Lastly, repeat for other end using diagram B using diagram A.

Result:

Thus, study of different type of wires, and working of own ethernet has been studied completely.